MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

AMPLIFIER, VIDEO, ANALOG, KEY CODE MFR

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic video amplifier module.

1.2 Classification. This specification covers two classes of modules.

Class I and class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787	-	Modules, Standard Electronic, General Specification for.
MIL-R-39035	-	Resistor, Variable, Nonwire Wound (Adjustment Type), Established Reliability, General Specification for

STANDARDS

MILITARY

MIL-STD-1389	-	Design Requirements for Standard Electronic Modules.
MIL-STD-1665	-	Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.585 failures per million operating hours for class I or 2.106 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be MFR.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/323-1 for class I and M28787/323-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.5.9 Potentiometers. Potentiometers selected for use in the module shall conform to the requirements of this document and shall be in accordance with MIL-R-39035, except as modified herein. Alternate potentiometers may be selected provided there is documented evidence that they are proven to be equivalent in all functional respects and that they conform to or exceed the reliability requirements of this document. The potentiometer lead screws shall be accessible from the module fin. Access holes shall be specified on figure 1. The lead screw adjustment of the potentiometer shall have a minimum of 10 mechanical turns.

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 2.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

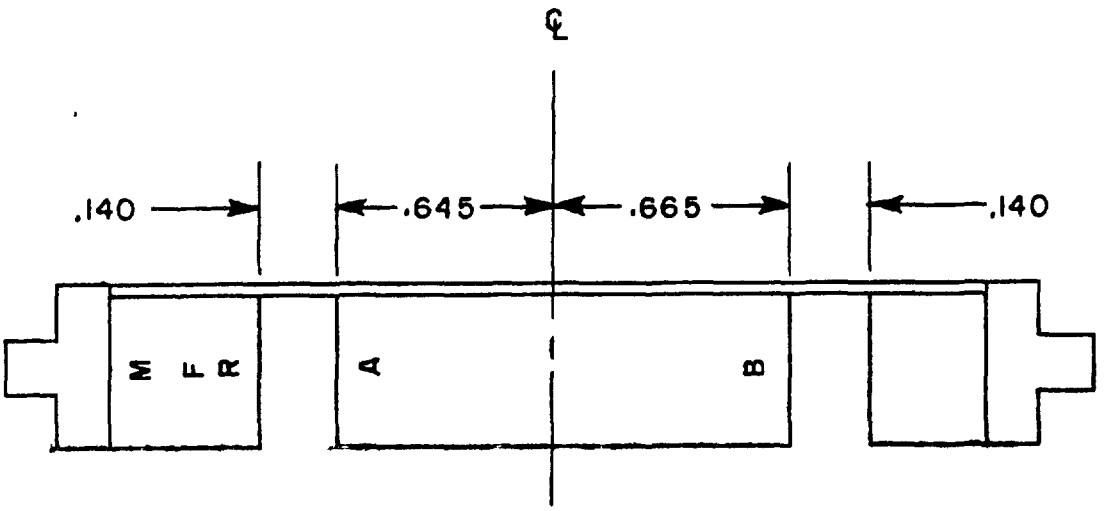
3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

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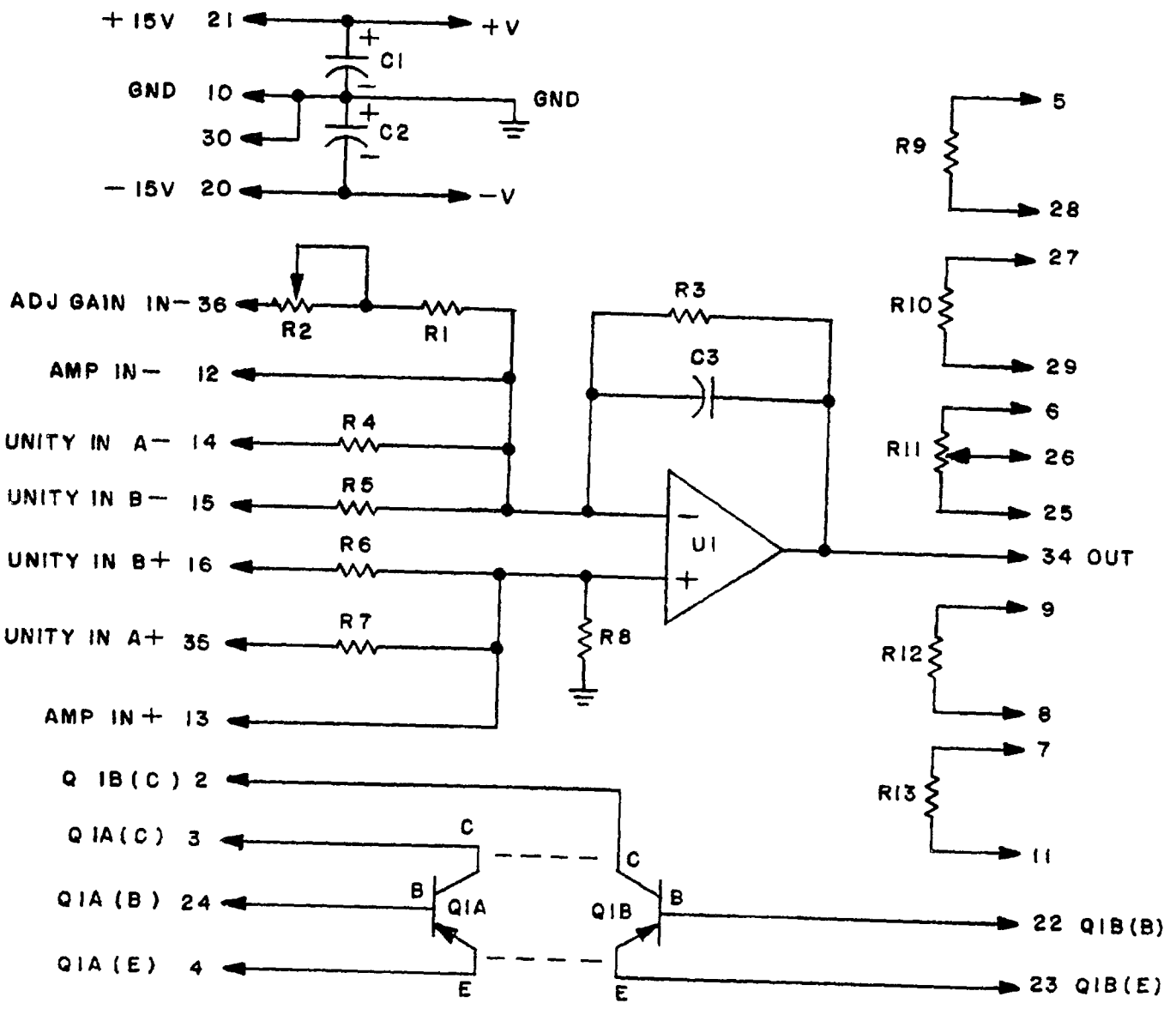


INCHES	MM
.140	3.56
.645	16.38
.665	16.89

NOTES:
1. The letters A and B shall be marked on the frame as shown adjacent to the potentiometer access holes. A shall designate R11 and B shall designate R2 as assigned on figure 2.
2. All dimensions are in inches with ±.005 (0.13 mm) tolerance.
3. Metric equivalents are given for general information only.

FIGURE 1. Location of potentiometer access holes.

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NOTES:
1. C1 = C2 = 1 µF.
2. C3 = 5.6 pF.
3. R1 = 220Ω ±1%, 1/8 W.
4. R2 = R11 = 5 kΩ, 1/8 W, potentiometer.
5. R3-R7, R9, R10 = 2.2 kΩ ±2%, 1/8 W.
6. R8 = 1 kΩ ±2%, 1/8 W.
7. R12 = R13 = 22Ω ±5%, 1/4 W.
8. Pins 1, 17, 18, 19, 31, 32, 33, 37, 38, 39, and 40 are unused.

FIGURE 2. MFR module function diagram, video amplifier.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A	V_S stress	II	Q	-120	120	-120	120	-120	120	-120	120	-120	120	mV
B1-B11	Isolation													
	Initial	III	X	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	III	X	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
C1	Power supply current (+V)	IV	X	0	30	0	30	0	30	0	30	0	30	mA
C2	Power supply current (-V)	IV	X	0	30	0	30	0	30	0	30	0	30	mA
D1,D2	Power supply filter capacitance	V	X	0.75	1.25	0.75	1.25	0.75	1.25	0.75	1.25	0.75	1.25	µF
E1	Resistance	VI	X	4710	5730	4710	5730	4710	5730	4710	5730	4710	5730	Ω
E2	Resistance	VI	X	215	245	215	245	215	245	215	245	215	245	Ω
E3,11	Resistance	VI	X	4500	5520	4500	5520	4500	5520	4500	5520	4500	5520	Ω
E4	Resistance	VI	X	0	20	0	20	0	20	0	20	0	20	Ω
E5-E10	Resistance	VI	X	2150	2250	2150	2250	2150	2250	2150	2250	2150	2250	Ω
E12,E13	Resistance	VI	X	20.5	23.5	20.5	23.5	20.5	23.5	20.5	23.5	20.5	23.5	Ω
E14	Resistance	VI	X	980	1020	980	1020	980	1020	980	1020	980	1020	Ω
E15,E16	Resistance	VI	X	0	1	0	1	0	1	0	1	0	1	Ω
F	Input offset voltage	VII	X	-75	75	-75	75	-75	75	-75	75	-75	75	mV
G	AC gain check	VIII	X	-0.45	0.43	-0.45	0.43	-0.45	0.43	-0.45	0.43	-0.45	0.43	dB
H	10 MHz gain check	IX	S	-3	3	-3	3	-3	3	-3	3	-3	3	dB
J1,J2	Output slew rate check	X	X	200	1000	200	1000	200	1000	200	1000	200	1000	V/µs
K	Common mode rejection ratio check	XI	X	-46	-20	-46	-20	-46	-20	-46	-20	-46	-20	dB
L	Distortion	XII	X	0	1	0	1	0	1	0	1	0	1	%
M	Noise	XIII	S	-56	0	-56	0	-56	0	-56	0	-56	0	dB

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
N1,N2	Forward current transfer ratio (dc) check (h_{FE})	XIV	X	300	900	250	900	300	900	175	900	300	900	mA/mA
N3	Beta match (h_{FE}) ratio	XIV	X	0.9	1.1	0.9	1.1	0.9	1.1	0.8	1.2	0.8	1.2	Ratio
P1,P2	Saturation voltage	XV	X	-250	0	-250	0	-250	0	-250	0	-250	0	mV
Q1,Q2	Breakdown voltage ($V_{(BR)EBO}$) check	XVI	S	---	-5	---	-5	---	-5	---	-5	---	-5	V
R1,R2	Breakdown voltage ($V_{(BR)CEO}$) check	XVII	S	---	-60	---	-60	---	-60	---	-60	---	-60	V
S	Output latch	Figure 5	Q	-750	750	-750	750	-750	750	-750	750	-750	750	mV

1/ "X" parameters shall be 100 percent tested during module acceptance test.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.
4/ Unless otherwise specified, limits are for initial and end-of-life.

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4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through R shall be performed in any order with the following exceptions. If test conditions A or B are part of a sequence of tests, the order of testing shall be as follows: A, B where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through R in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions C through R shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 5. Following the life test and prior to disconnecting the power, electrical test S shall be performed in accordance with figure 5. While still at the maximum class temperature, electrical tests C through R shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests B through R shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

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6.4 Module description. The following electrical parameters reflect the 0°C to 60°C and -55°C to 85°C end-of-life limits from table I.

	Class I	Class II	Unit
Maximum power dissipation	1.43	1.43	W
Output voltage swing	±10 min	±10 min	Vp-p
Bandwidth (small signal)	10 min	10 min	MHz
Analog input level	±18 max	±18 max	V

6.5 Module function. This module contains a multiple input high speed amplifier for video and other applications. Inverting and noninverting modes with gain range from 0 to 20 dB are available. Adjustments are accessible through the module fin as shown on figure 1. Miscellaneous components are as shown on figure 2.

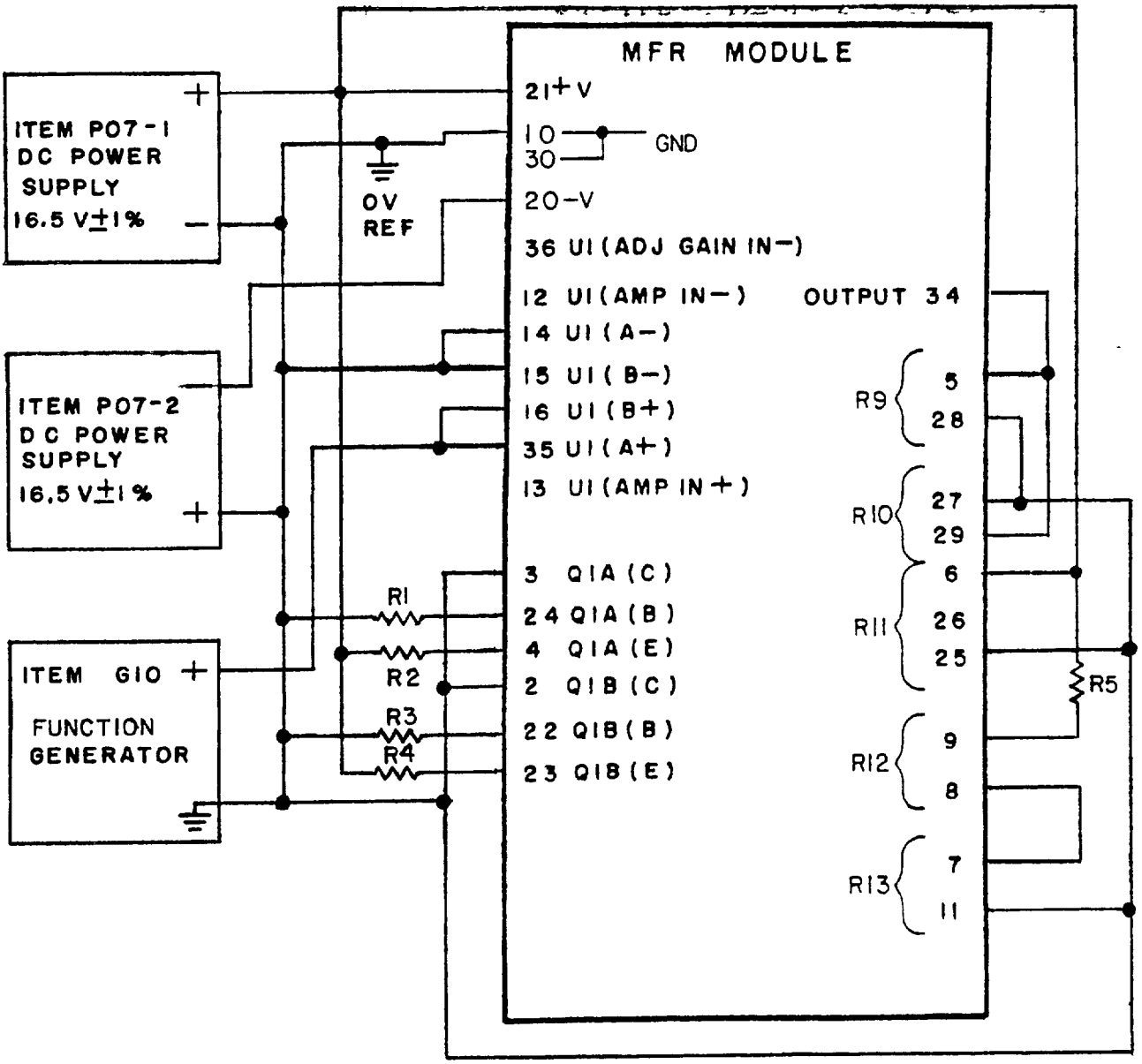
6.5.1 Maximum ratings. The maximum ratings for supply voltages and input levels given in 6.4 represent extremes beyond which damage to the module may occur. Operation at the extremes is performed in this specification to insure that the module will survive these levels without damage. Performance specifications are tested at supply levels 8 percent less than maximum ratings and input levels equal to or between the supply voltages.

Agent:
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NOTES:
1. R1 = R3 = 150 kΩ ±1%, 1/8 W.
2. R2 = R4 = 1.5 kΩ ±1%, 1/4 W.
3. R5 = 180Ω ±5%, 2 W.
4. Item G10 setup:
   Amplitude = 2.3 V rms ±1%.
   Frequency = 5 kHz ±1%.
   Sine wave.

FIGURE 3. Operating temperature and component temperature test circuit.

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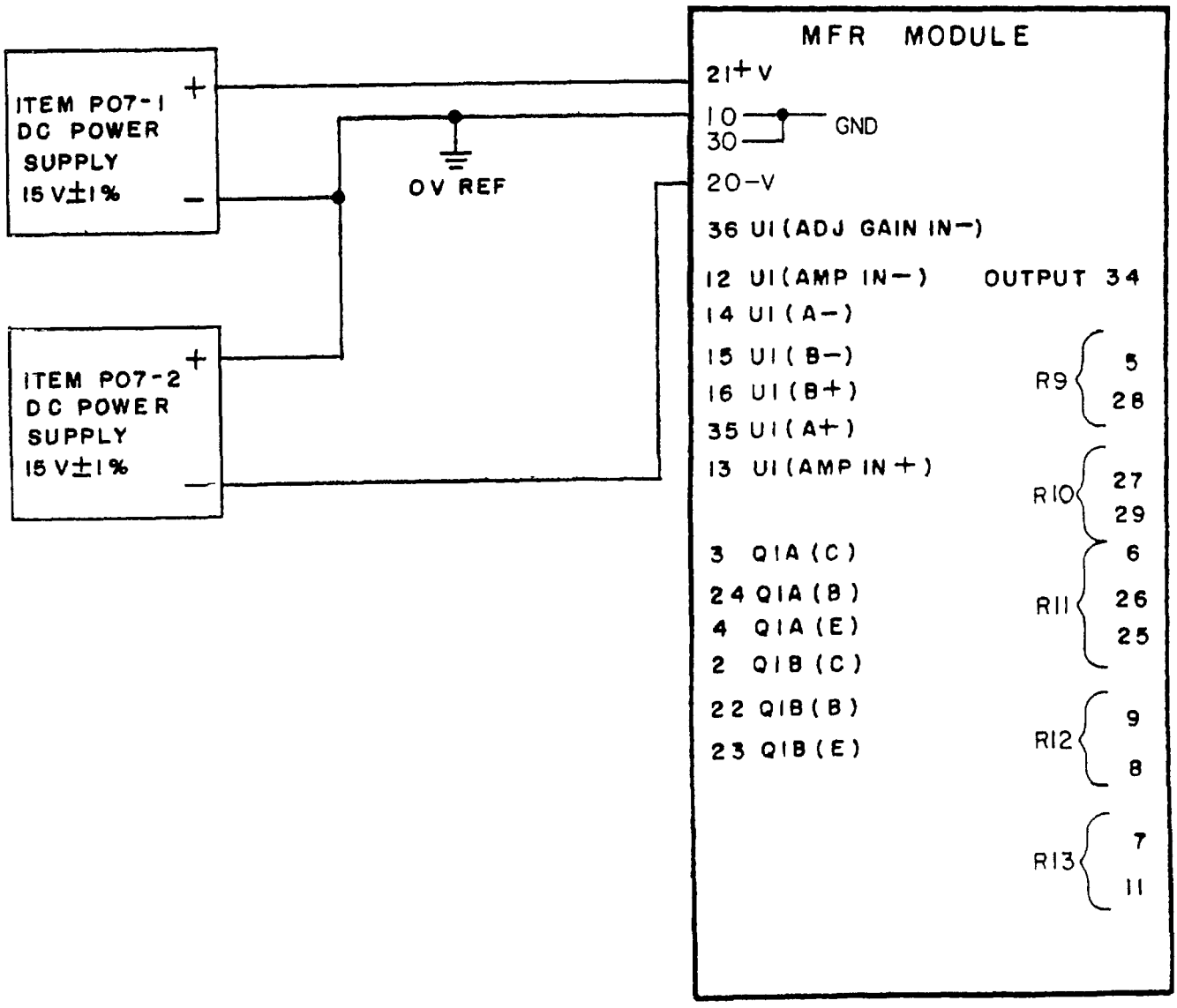
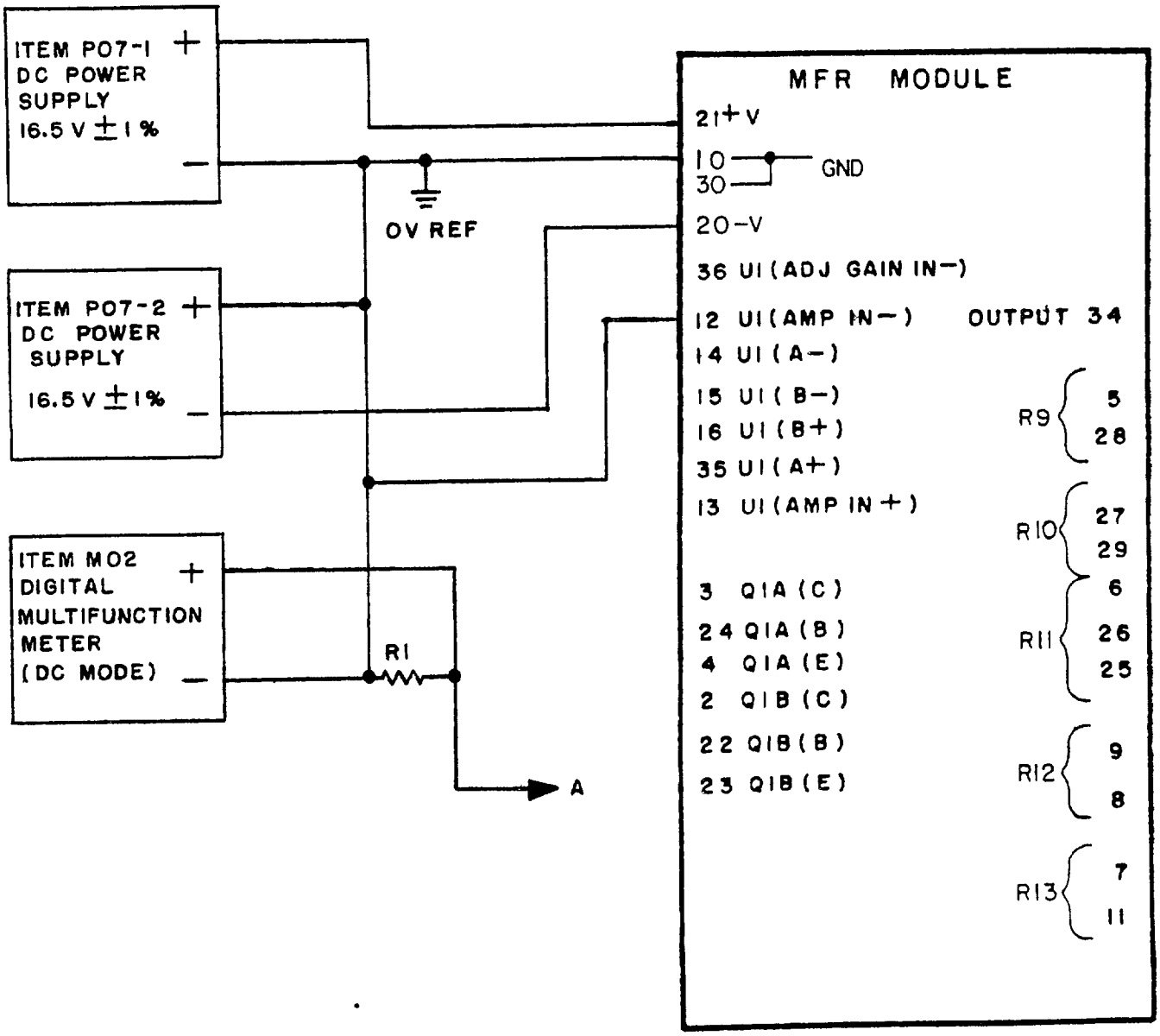


FIGURE 4. Vibration test circuit.

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NOTES:
1. R1 = 1 kΩ ±5%, 1/4 W.
2. Following the life test period prior to disconnecting the power, connect point A to pin 34. The voltage shall meet the limits specified in table I for test condition S.

FIGURE 5. Life test circuit.

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TABLE II. V_S stress test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
A	V_S stress	-120	120	mV	6

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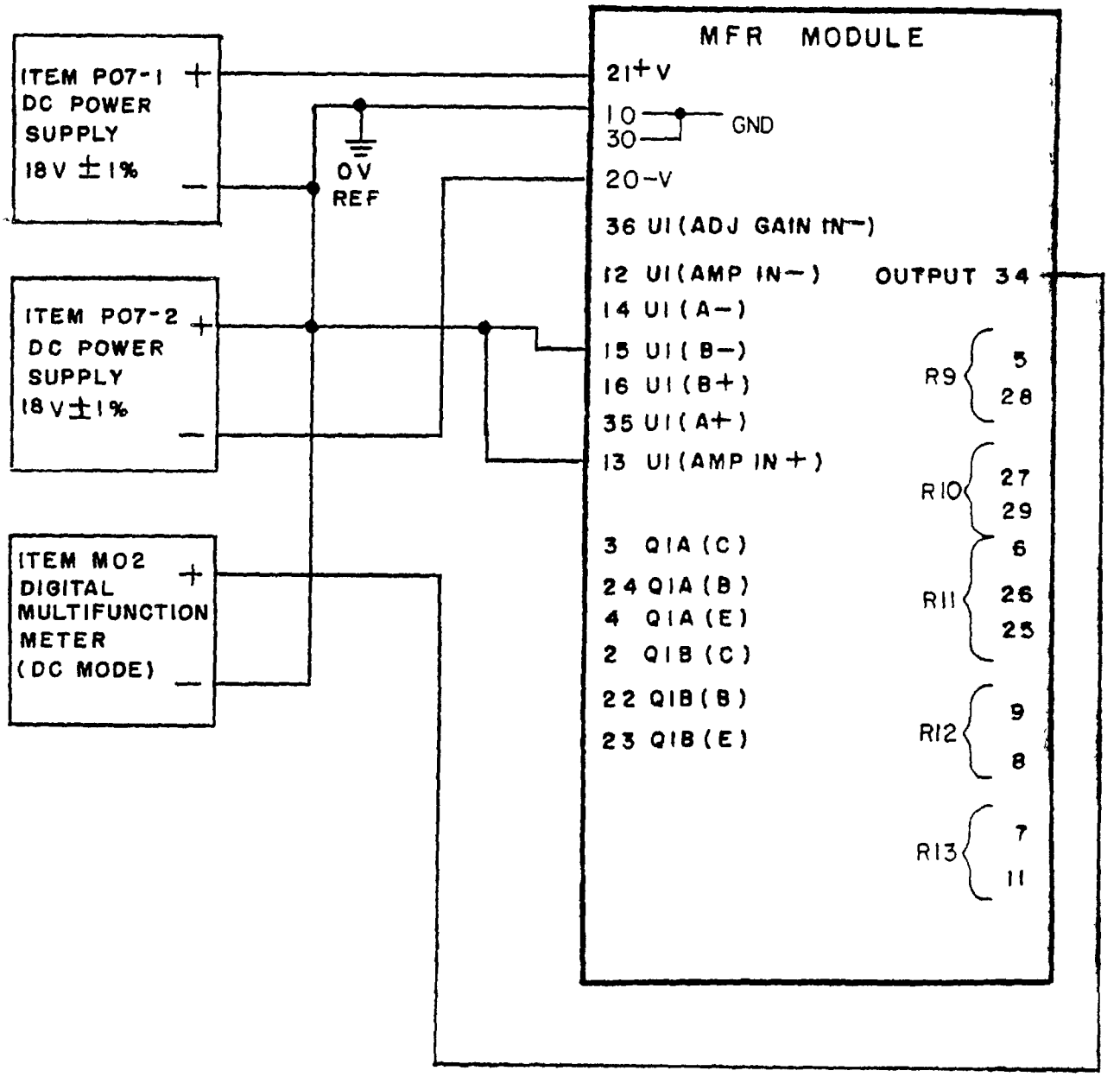


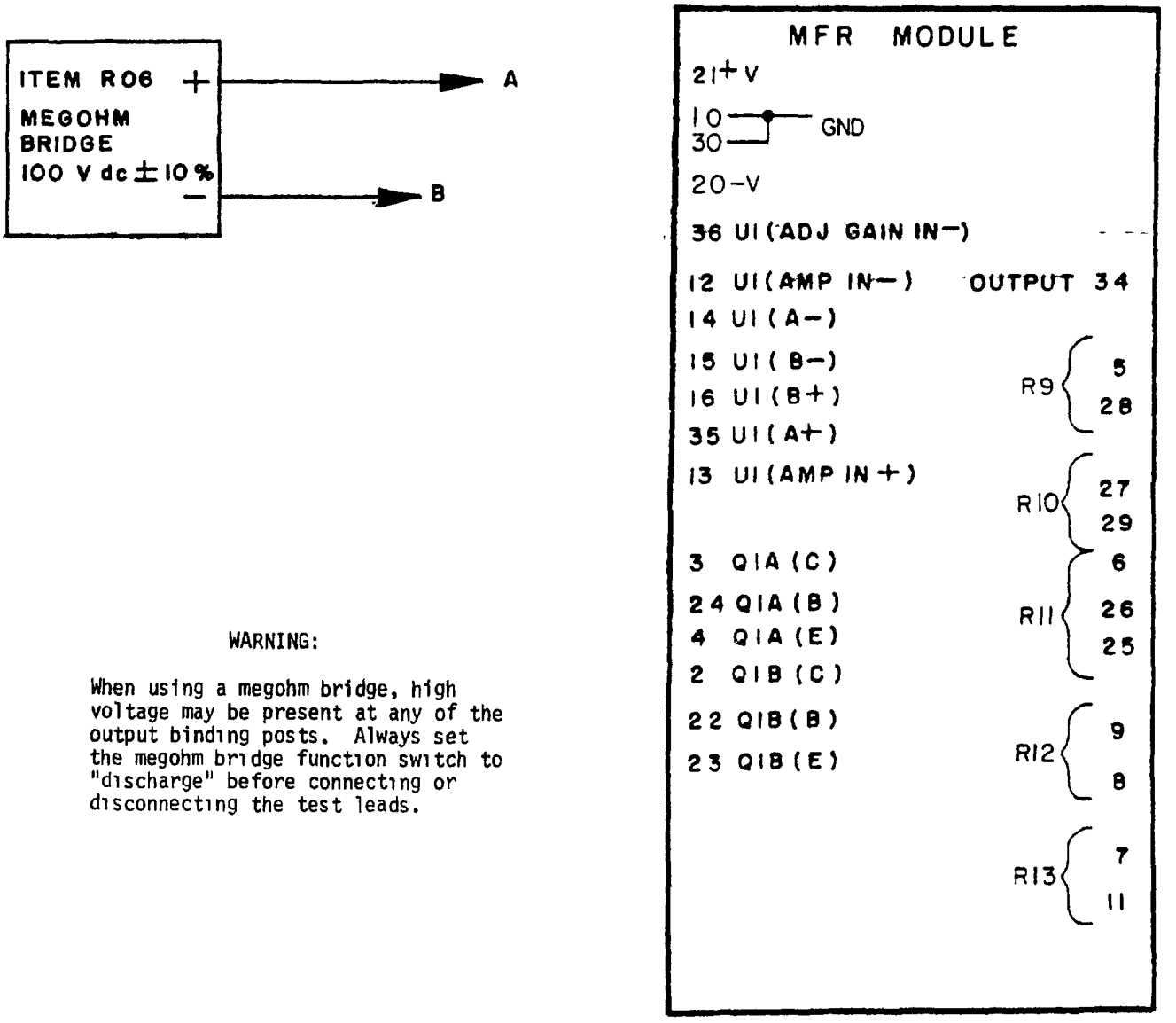
FIGURE 6. V_S stress test circuit for table II.

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TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	Isolation, unused to unused pins	17,18,19,31,32, 33,37,38,39,40	1	10	∞	MΩ	7
B2	"	18,19,31,32,33, 37,38,39,40	17	"	"	"	"
B3	"	19,31,32,33,37, 38,39,40	18	"	"	"	"
B4	"	31,32,33,37,38, 39,40	19	"	"	"	"
B5	"	32,33,37,38,39, 40	31	"	"	"	"
B6	"	33,37,38,39,40	32	"	"	"	"
B7	"	37,38,39,40	33	"	"	"	"
B8	"	38,39,40	37	"	"	"	"
B9	"	39,40	38	"	"	"	"
B10	"	40	39	"	"	"	"
B11	Isolation, used to unused pins	2,3,4,5,6,7,8, 9,10,11,12,13, 14,15,16,20,21, 22,23,24,25,26, 27,28,29,30,34, 35,36	1,17,18,19,31, 32,33,37,38, 39,40	"	"	"	"

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WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

FIGURE 7. Isolation test circuit for table III.

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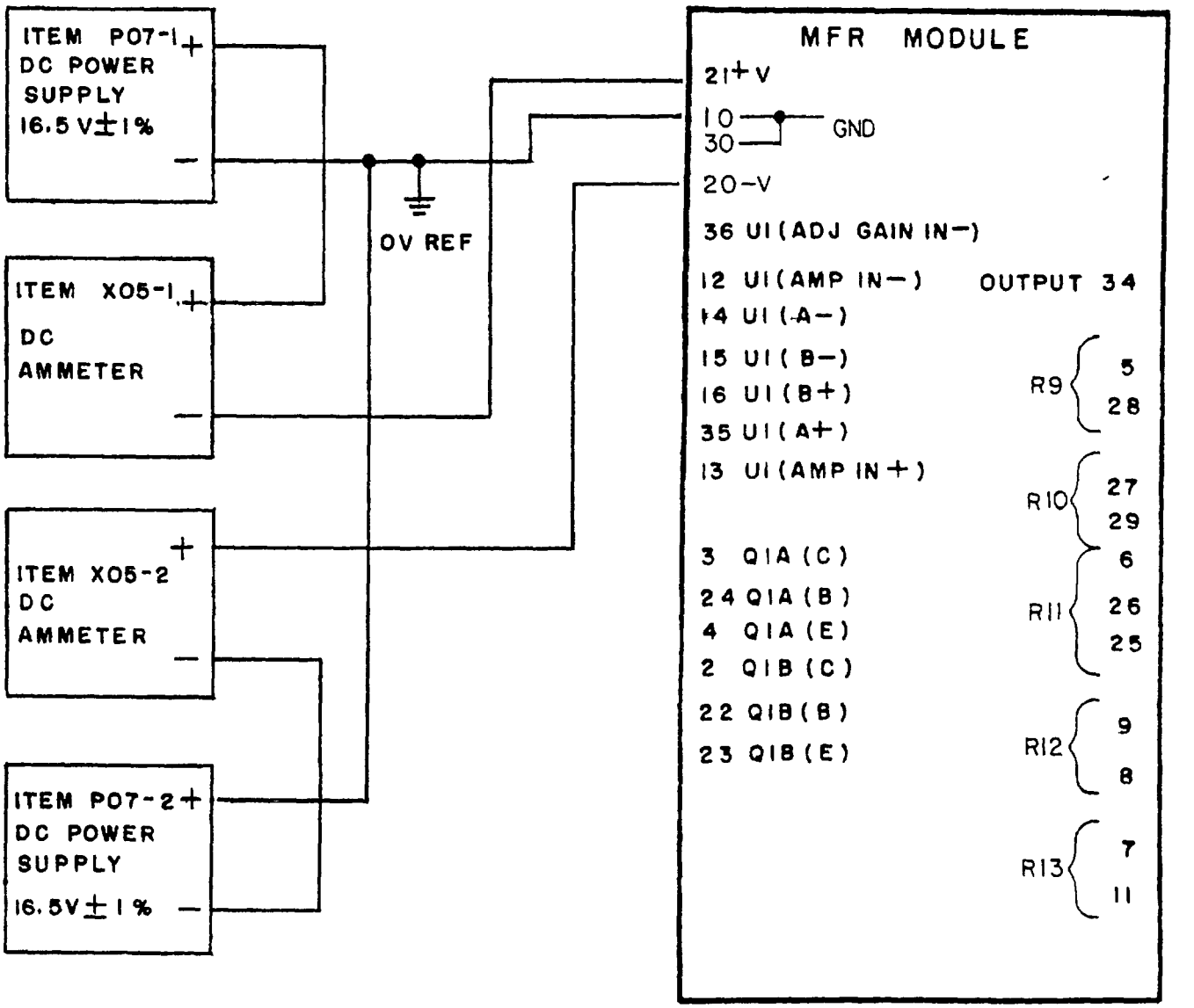
TABLE IV. Power supply current test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
C1	Power supply current (+V)	0	30	mA	8
C2	Power supply current (-V)	0	30	mA	8

TABLE V. Power supply filter capacitance test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
D1	Power supply filter capacitance	21	10	0.75	1.25	µF	9
D2	Power supply filter capacitance	10	20	0.75	1.25	µF	9

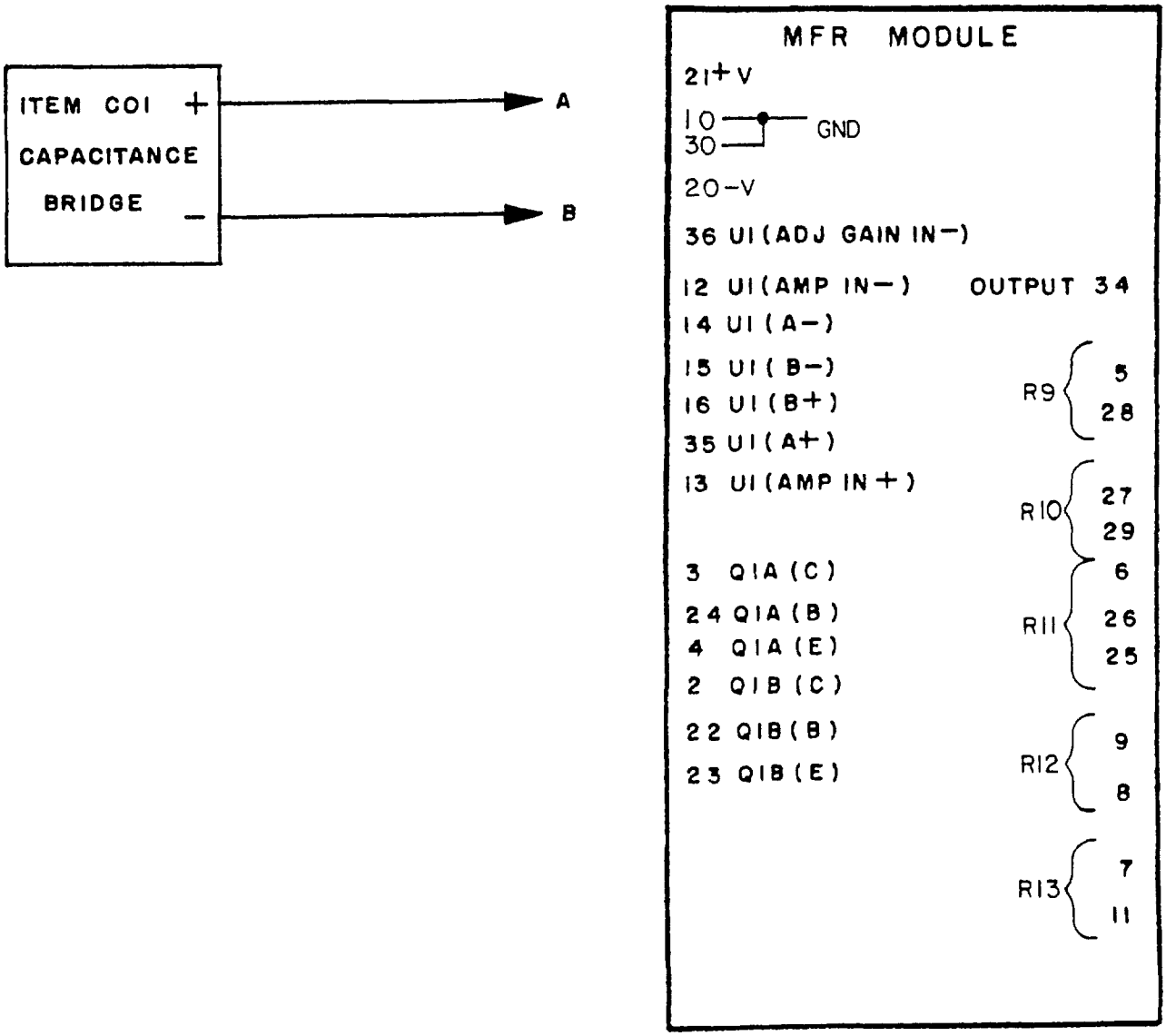
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NOTES:
1. Since items X05-1 and X05-2 are not used simultaneously, only one item X05 is required.
2. Read item X05-1 for test condition C1 and item X05-2 for test condition C2.

FIGURE 8. Power supply current test circuit for table IV.

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NOTE: Item C01 setup:
Frequency = 120 Hz.
Vpk = 0.3 V maximum.

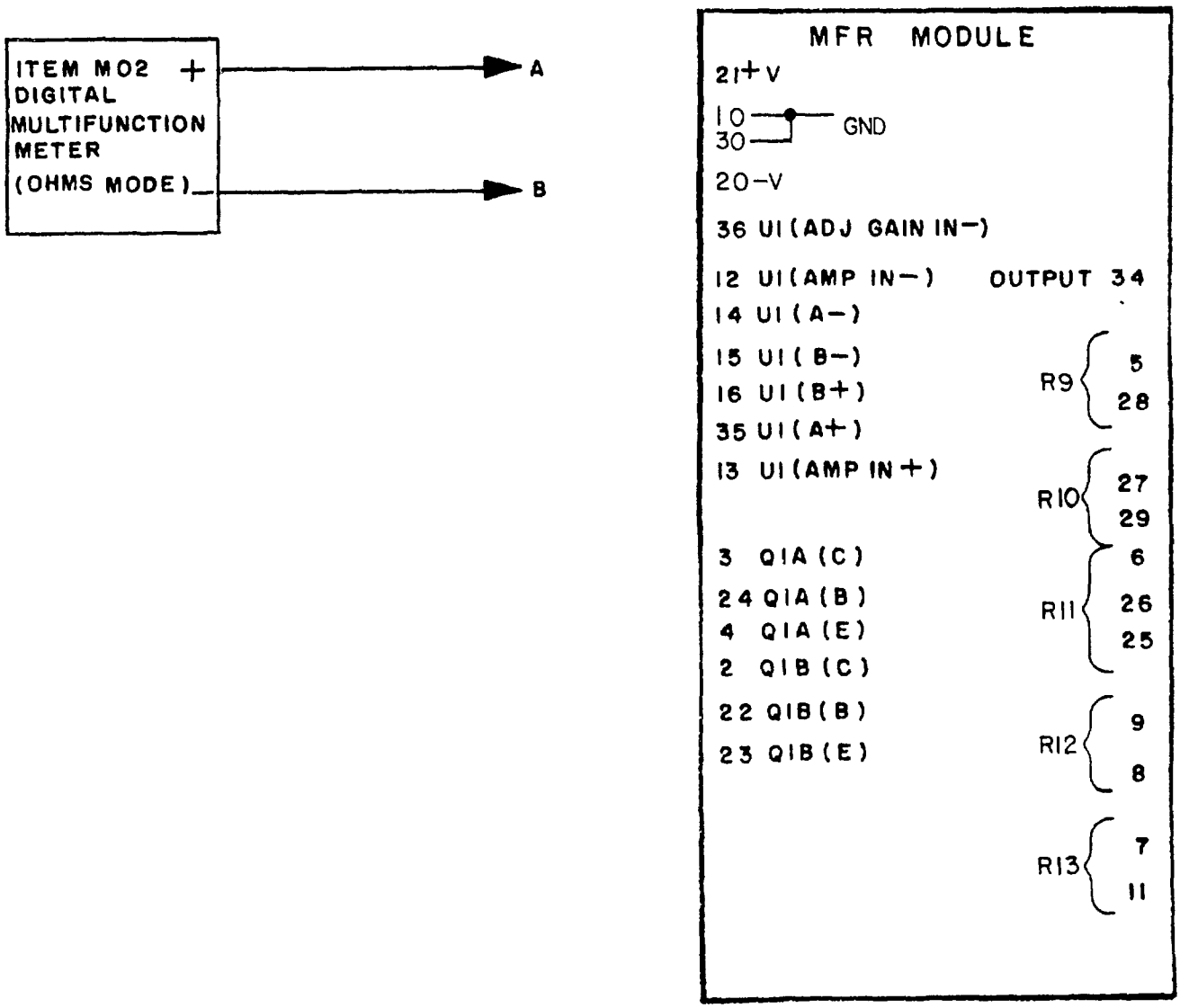
FIGURE 9. Power supply filter capacitance test circuit for table V.

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TABLE VI. Resistance test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
E1	R2 + R1 Resistance	36	12	4710	5730	Ω	10
E2	R1 "	36	12	215	245	"	"
E3	R11 "	26	25	4500	5520	"	"
E4	R11 "	26	25	0	20	"	"
E5	R4 "	12	14	2150	2250	"	"
E6	R5 "	12	15	"	"	"	"
E7	R6 "	16	13	"	"	"	"
E8	R7 "	35	13	"	"	"	"
E9	R9 "	5	28	"	"	"	"
E10	R10 "	27	29	"	"	"	"
E11	R11 "	6	25	4500	5520	"	"
E12	R12 "	9	8	20.5	23.5	"	"
E13	R13 "	7	11	20.5	23.5	"	"
E14	R8 "	13	10	980	1020	"	"
E15	"	10	30	0	1	"	"
E16	"	10	Frame	0	1	"	"

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NOTES: 1. Adjust R2(B) and R11(A) as shown below:

Test condition	R2 (B) 1/	R11 (A) 1/
E1	CW	--
E2	CCW	--
E3	--	CCW
E4	--	CW

1/ CW = Full clockwise position.
CCW = Full counterclockwise position.

2. For test condition E16, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

FIGURE 10. Resistance test circuit for table VI.

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TABLE VII. Input offset voltage test.

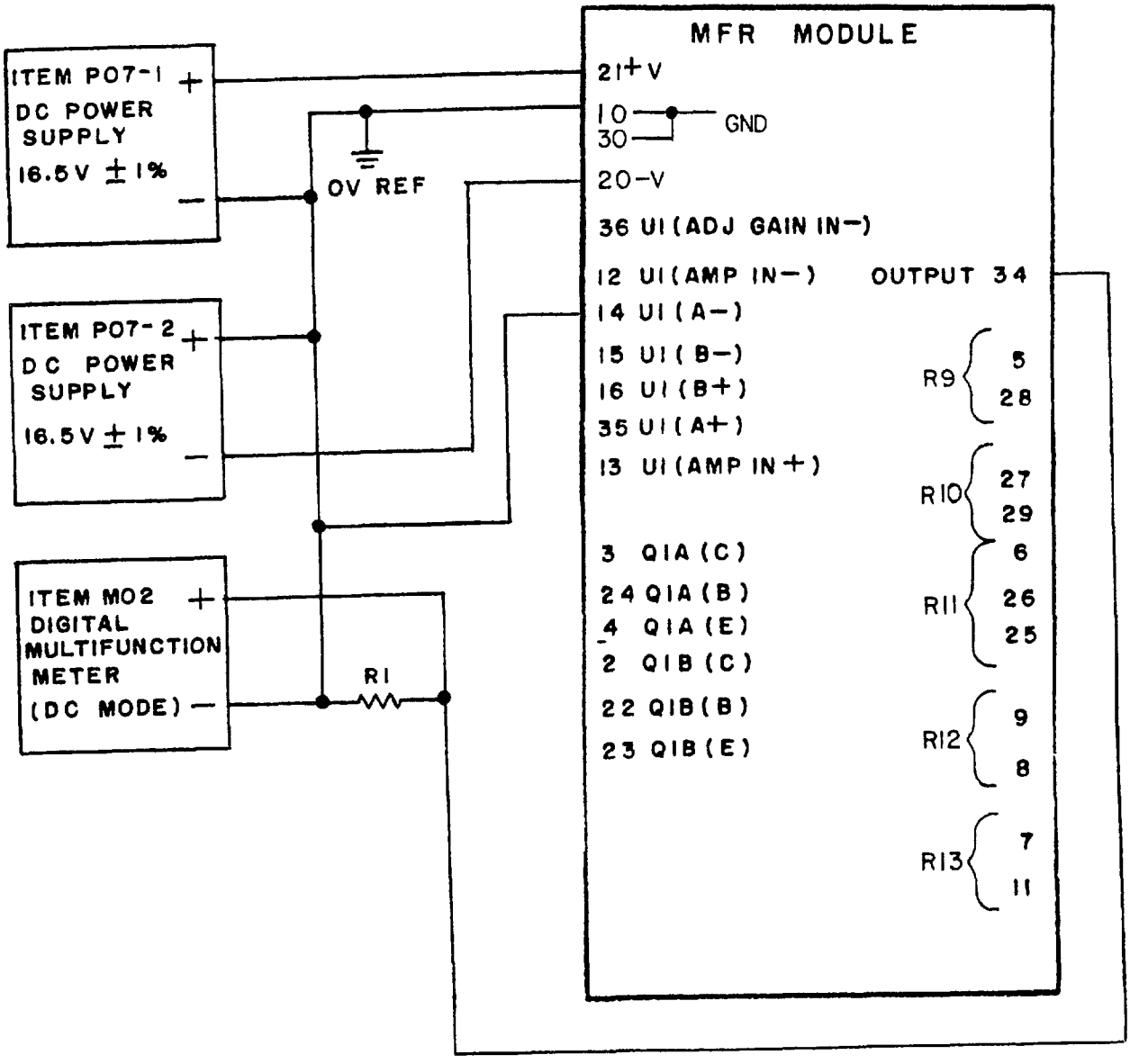
Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
F	Input offset voltage	-75	75	mV	11

TABLE VIII. AC gain check test. 1/

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
G	AC gain check	0.95	1.05	V rms	12

1/ AC gain (table I, dB) = 20 log $\frac{V_{OUT}}{V_{IN}}$ = 20 log $\frac{\text{Item W07 reading}}{1 \text{ V rms}}$

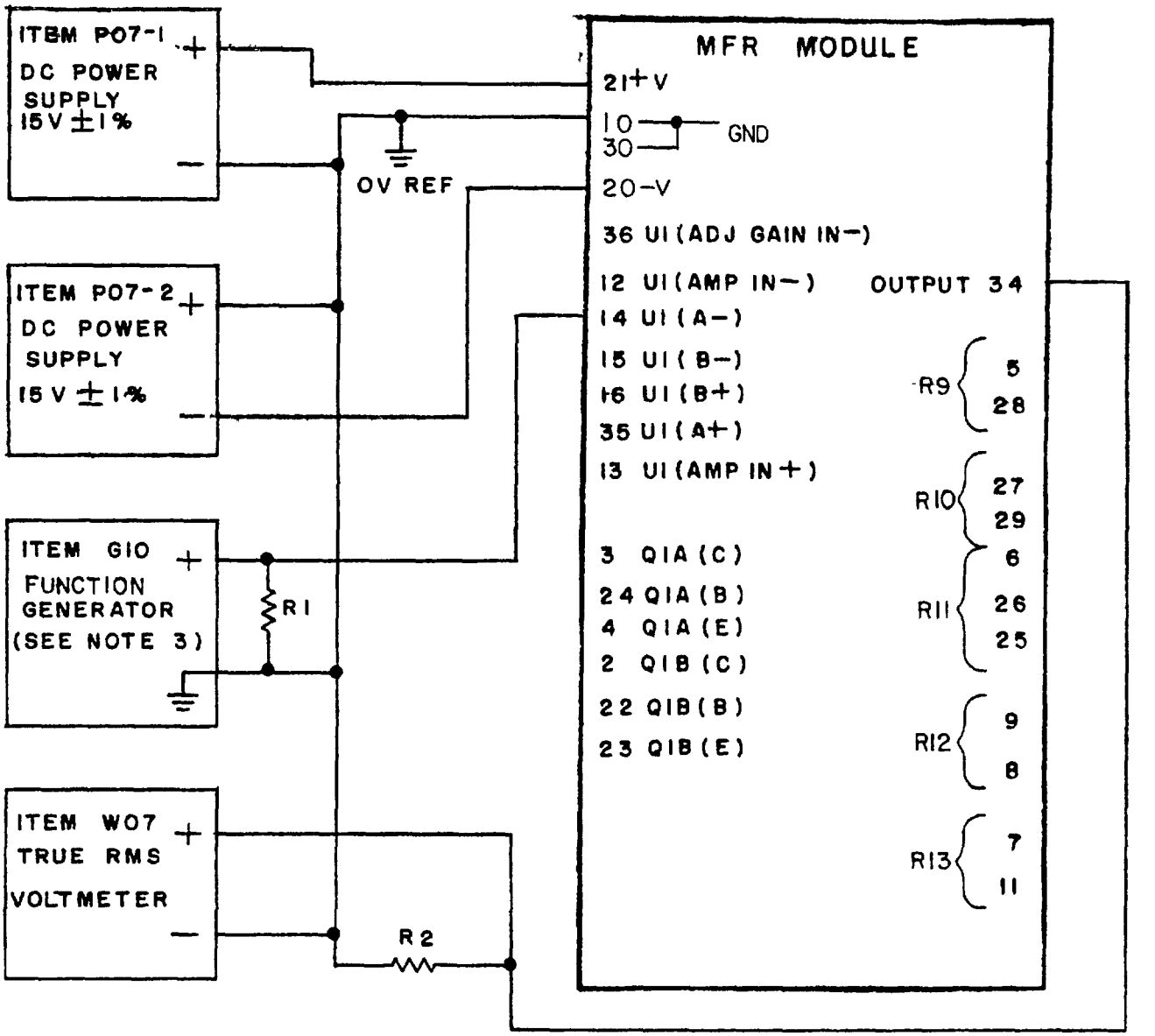
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NOTE: R1 = 1 kΩ ±5%, 1/10 W.

FIGURE 11. Input offset voltage test circuit for table VII.

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NOTES:
1. R1 = 51Ω ±5%, 1 W.
2. R2 = 1 kΩ ±5%, 1/8 W.
3. Item G10 setup:
   Amplitude = 1 V rms ±1%.
   Frequency = 1 kHz ±1%.
   Sine wave.

FIGURE 12. AC gain check test circuit for table VIII.

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TABLE IX. 10 MHz gain check test. 1/ 2/

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
H	10 MHz gain check	0.707	1.414	V rms	13

1/ This test verifies a minimum bandwidth of 10 MHz.

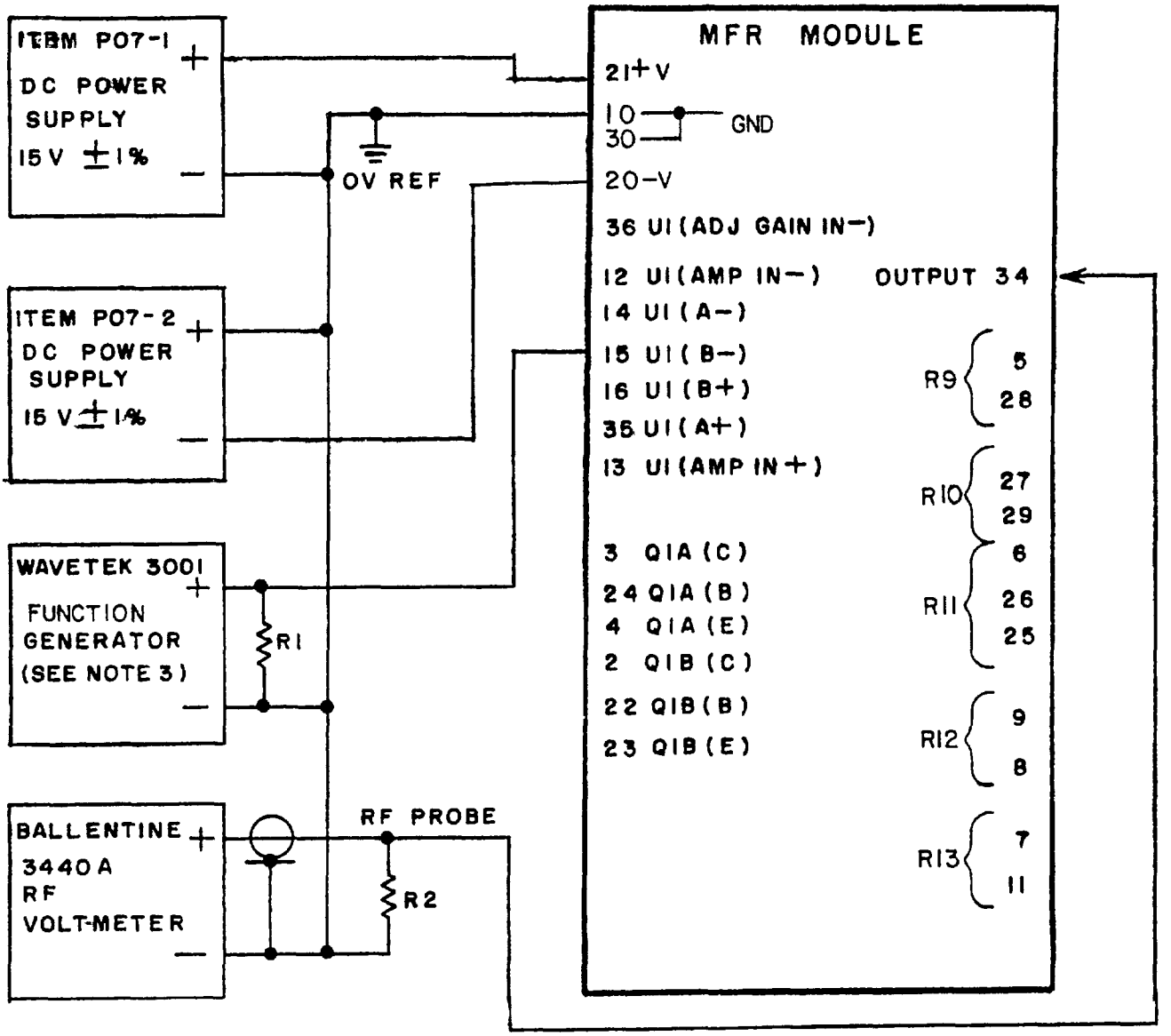
2/ 10 MHz gain (table I, dB) = $20 \log \left(\frac{\text{RF voltmeter (Ballentine 3440A)}}{1 \text{ V rms}} \right)$

TABLE X. Output slew rate check test. 1/

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
J1	Output slew rate check	0.004	0.020	µs	14,15
J2	Output slew rate check	0.004	0.020	µs	14,15

1/ Output slew rate (table 1, V/µs) = $\frac{4V}{\Delta t \ (\mu s)}$

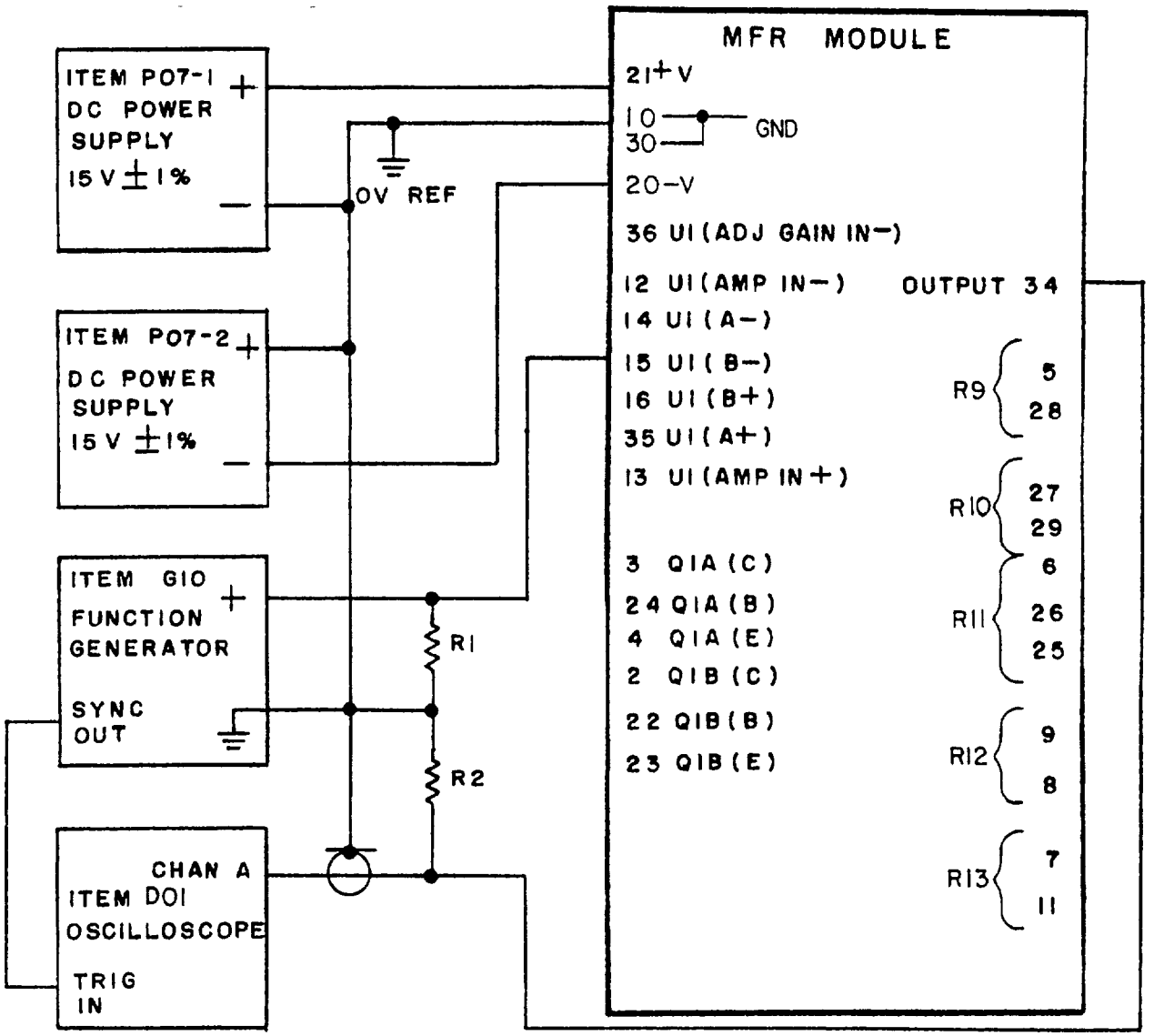
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NOTES:
1. R1 = 51Ω ±5%, 1 W.
2. R2 = 1 kΩ ±5%, 1/8 W.
3. Adjust sine wave generator (Wavetek 3001 or equivalent) to a 1 V rms ±1%, 10 MHz ±1%, reading at pin 15 using RF voltmeter (Ballentine 3440A or equivalent).
4. Use RF voltmeter (Ballentine 3440A or equivalent) with RF probe to measure output voltage.

FIGURE 13. 10 MHz gain check test circuit for table IX.

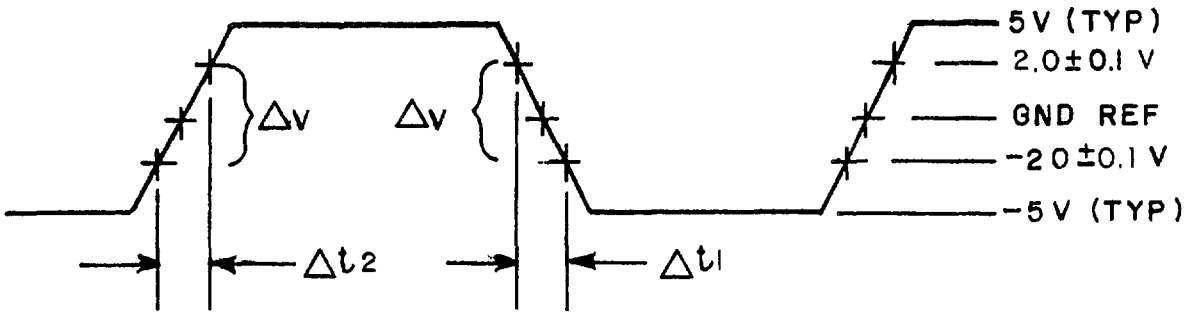
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NOTES:
1. R1 = 51Ω ±5%, 1 W.
2. R2 = 1 kΩ ±5%, 1/8 W.
3. Item G10 setup:
   Amplitude = 10 V p-p ±5%.
   Frequency = 100 kHz ±5%.

FIGURE 14. Output slew rate check test circuit for table X.

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NOTES:
1. ΔV/Δt = Output slew rate.
2. Measure Δt1, for test condition J1 and Δt2 for test condition J2.

FIGURE 15. Output slew rate check test circuit waveform for table X.

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TABLE XI. Common mode rejection ratio check test. 1/

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
K	Common mode rejection ratio check	5	100	mV rms	16

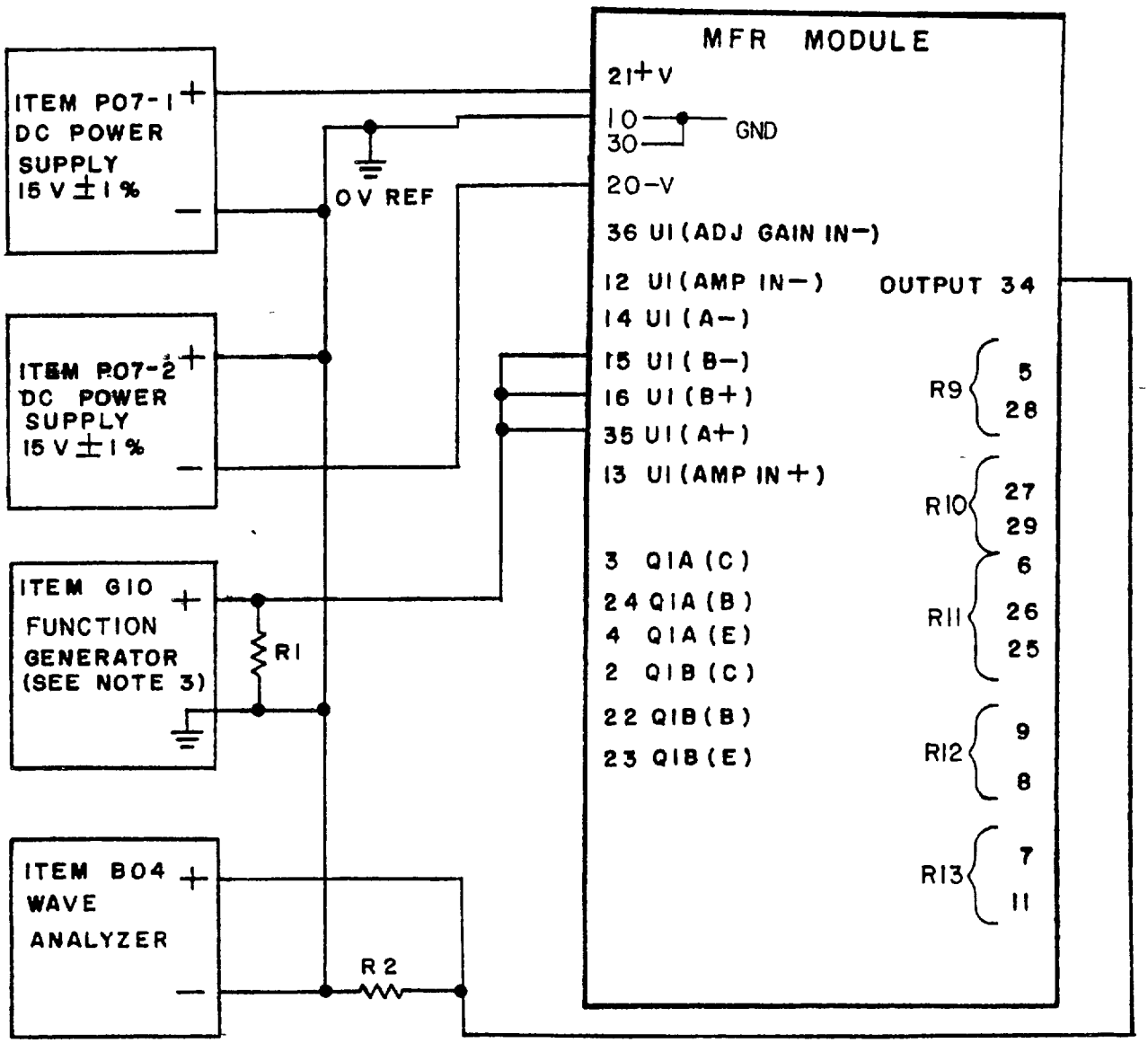
1/ Common mode rejection ratio (table I, dB) = $20 \log \left(\frac{V_{OUT}/Gain}{V_{IN}} \right)$ = $20 \log \left(\frac{\text{Item B04 reading}}{1 \text{ V rms}} \right)$

TABLE XII. Distortion test. 1/

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
L	Distortion	0	1	%	17

1/ This test also verifies a minimum output voltage swing of 10 Vp-p.

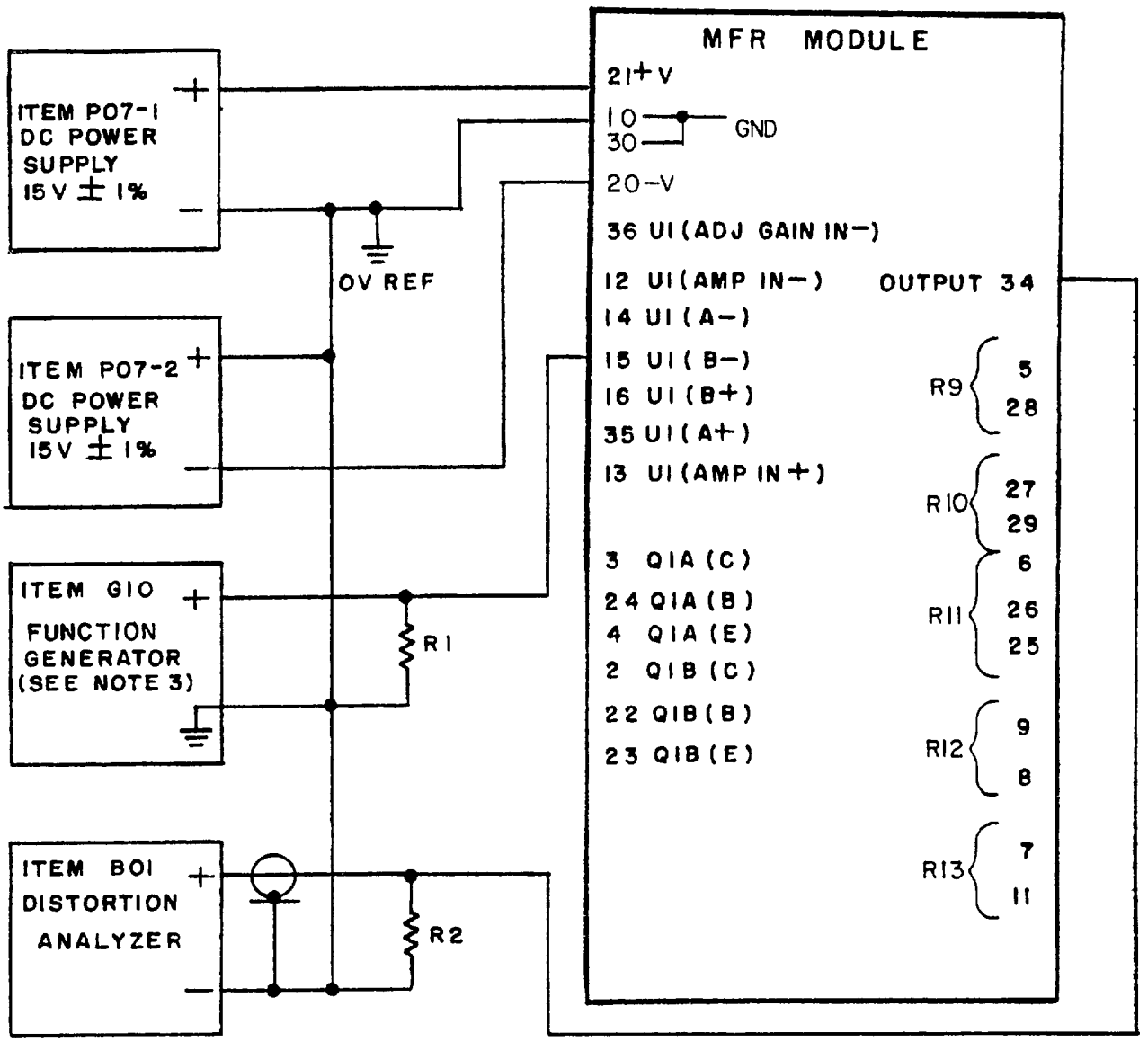
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NOTES:
1. R1 = 51Ω ±5%, 1 W.
2. R2 = 1 kΩ ±5%, 1/8 W.
3. Item G10 setup:
   Amplitude = 1 V rms ±1%.
   Frequency = 1 kHz ±1%.
   Sine wave.

FIGURE 16. Common mode rejection ratio check test circuit for table XI.

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NOTES:
1. R1 = 51Ω ±5%, 1 W.
2. R2 = 1 kΩ ±5%, 1/8 W.
3. Item G10 setup: –
   Amplitude = 3.54 V rms ±1%.
   Frequency = 1 kHz ±1%.
   Sine wave.

FIGURE 17. Distortion test circuit for table XII.

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TABLE XIII. Noise test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
M	Noise	-56	0	dB	18

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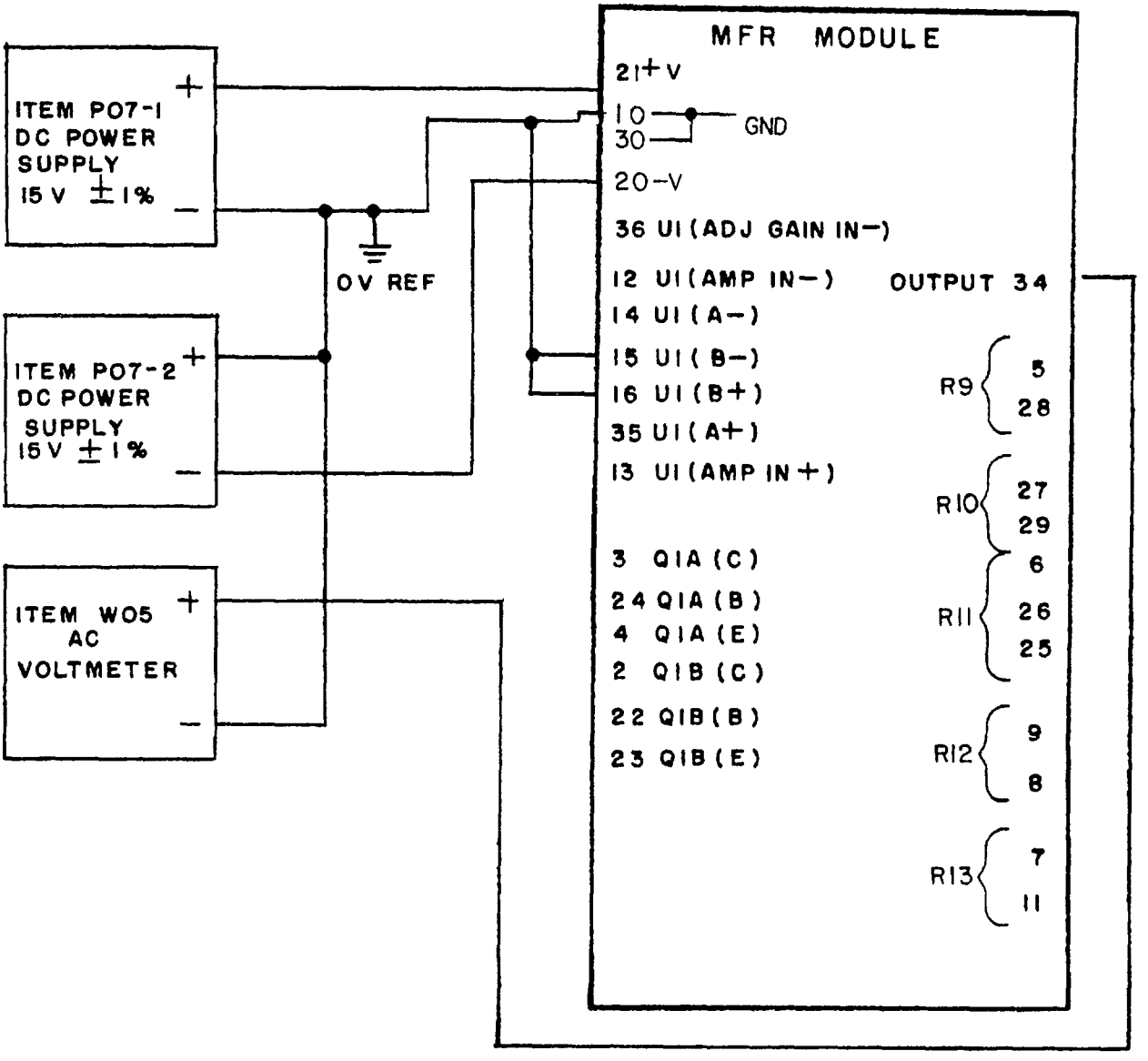


FIGURE 18. Noise test circuit for table XIII.

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TABLE XIV. Forward current transfer ratio (dc) (h_{FE}) check and beta match (h_{FE}) ratio test. 1/ 2/

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
N1	Q1A Forward current transfer ratio (dc) check (h_{FE}), I_C = -1 mA	4	24	3	1.1	3.33	µA	19
N2	Q1B Forward current transfer ratio (dc) check (h_{FE}), I_C = -1 mA	23	22	2	1.1	3.33	µA	19
N3	Beta match (h_{FE}) ratio	---	---	---	0.9	1.1	---	---

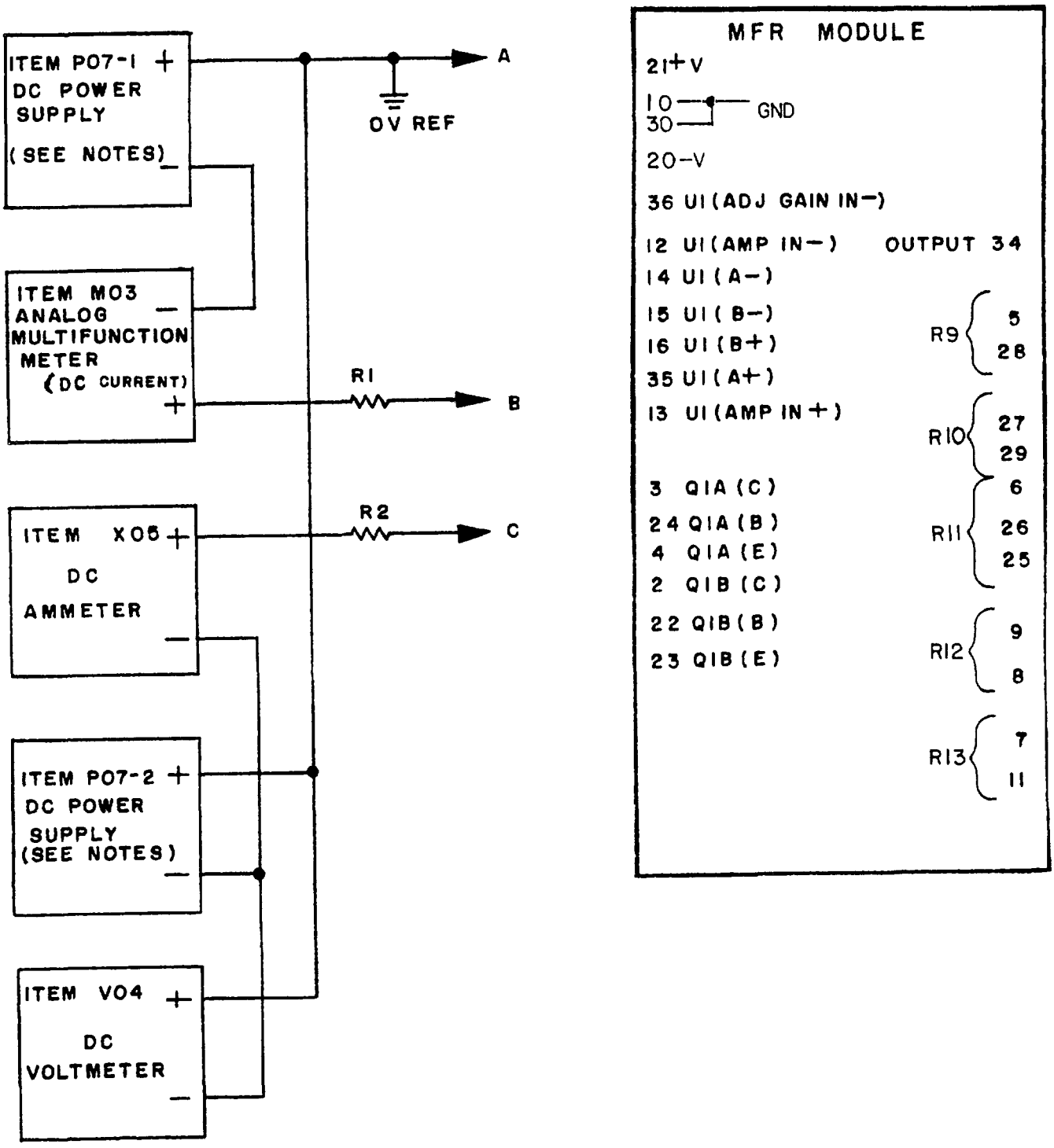
1/ Forward current transfer ratio (dc) (table I, mA/mA) = $\frac{1 \text{ mA}}{\text{Item M03 reading}}$

2/ Test condition N3 is calculated as follows: Test condition N3 (table I, ratio) = $\frac{\text{Test condition N1.}}{\text{Test condition N2}}$

TABLE XV. Saturation voltage test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
P1	Q1A Saturation voltage	24	3	-250	0	mV	20
P2	Q1B Saturation voltage	22	2	-250	0	mV	20

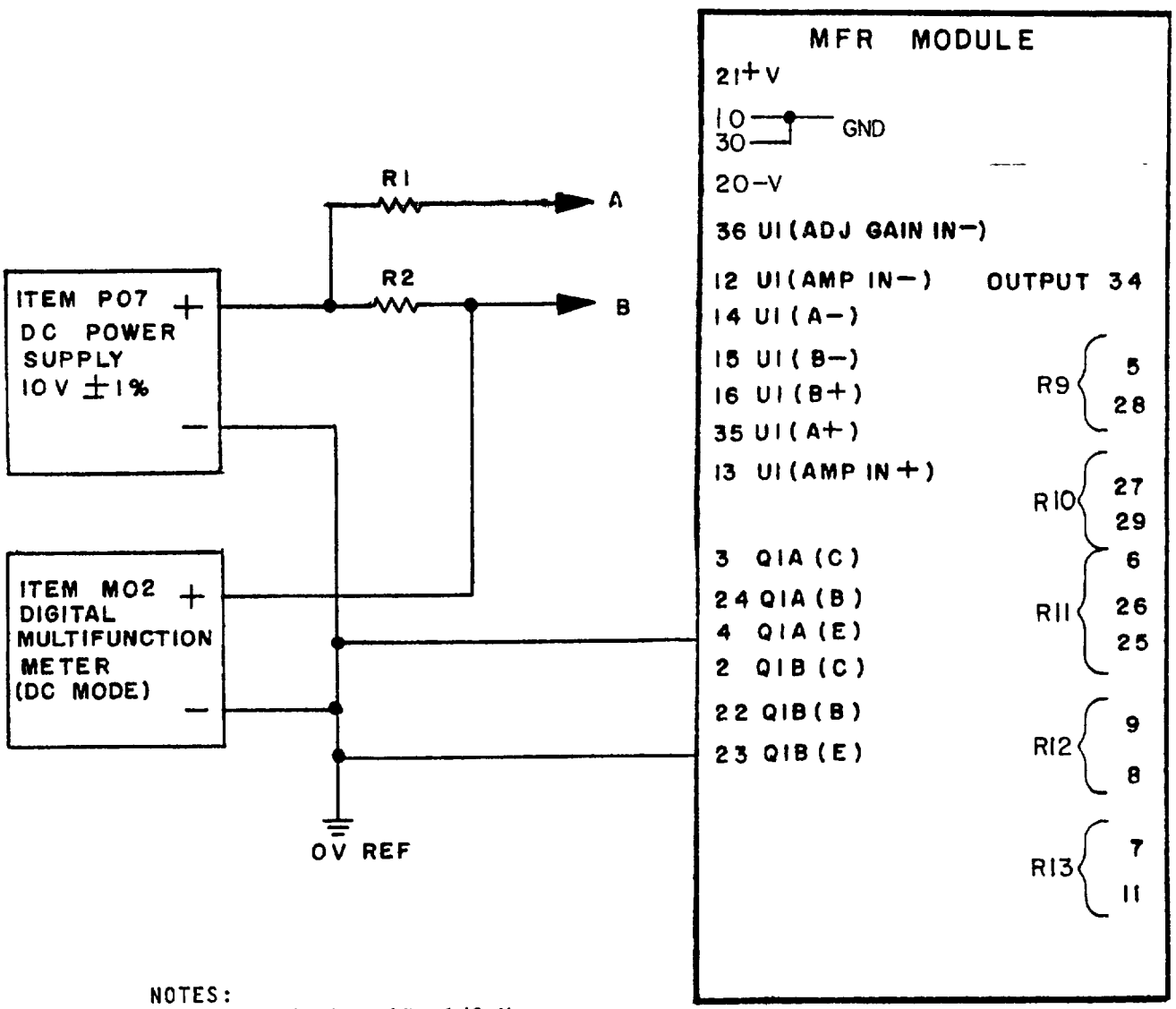
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NOTES:
1. R1 = 1 MΩ ±1%, 1/8 W.
2. R2 = 10Ω ±1%, 1/4 W.
3. The test shall be performed as follows:
   a. Initially set the power supplies to 0 V.
   b. Make connections as shown in table XIV.
   c. Adjust item P07-2 first for 5 V ±1% as measured on item V04.
   d. Adjust item P07-1 for a collector current of 1 mA ±1% as indicated on X05.
   e. The current measured on item M03 shall meet the limits of table XIV.
4. Test leads shall be used which minimize stray capacitance to pervent transistor saturation. Power supplies shall not have chassis ground terminals connected.

FIGURE 19. Forward current transfer ratio (dc) check (h_{FE}) and beta match (h_{FE}) ratio test circuit for table XIV.

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NOTES:
1. R1 = 100 kΩ ±1%, 1/8 W.
2. R2 = 10 kΩ ±1%, 1/8 W.

FIGURE 20. Saturation voltage test circuit for table XV.

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TABLE XVI. Breakdown voltage ($V_{(BR)EBO}$) check test. 1/

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
Q1	Q1A Breakdown voltage ($V_{(BR)\ EBO}$) check	24	0	10	µA	21
Q2	Q1B Breakdown voltage ($V_{(BR)\ EBO}$) check	22	0	10	µA	21

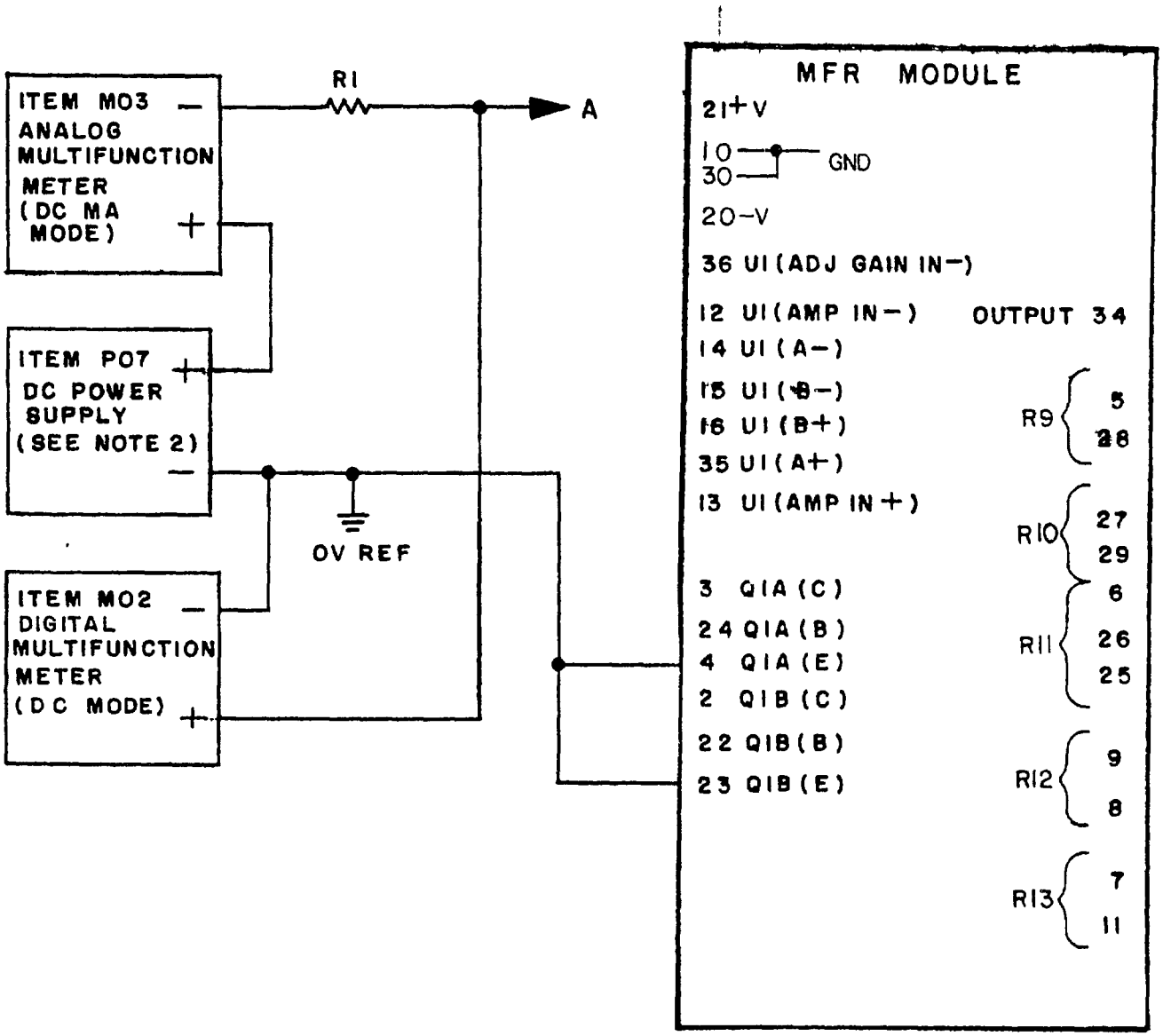
1/ This test guarantees a minimum $V_{(BR)EBO}$ of -5 V(table I).

TABLE XVII. Breakdown voltage ($V_{(BR)CEO}$) check test. 1/

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
R1	Q1A Breakdown voltage ($V_{(BR)\ CEO}$) check	3	-1	0	V	22
R2	Q1B Breakdown voltage ($V_{(BR)\ CEO}$) check	2	-1	0	V	22

1/ This test verifies a minimum $V_{(BR)CEO}$ of -60 V (table I).

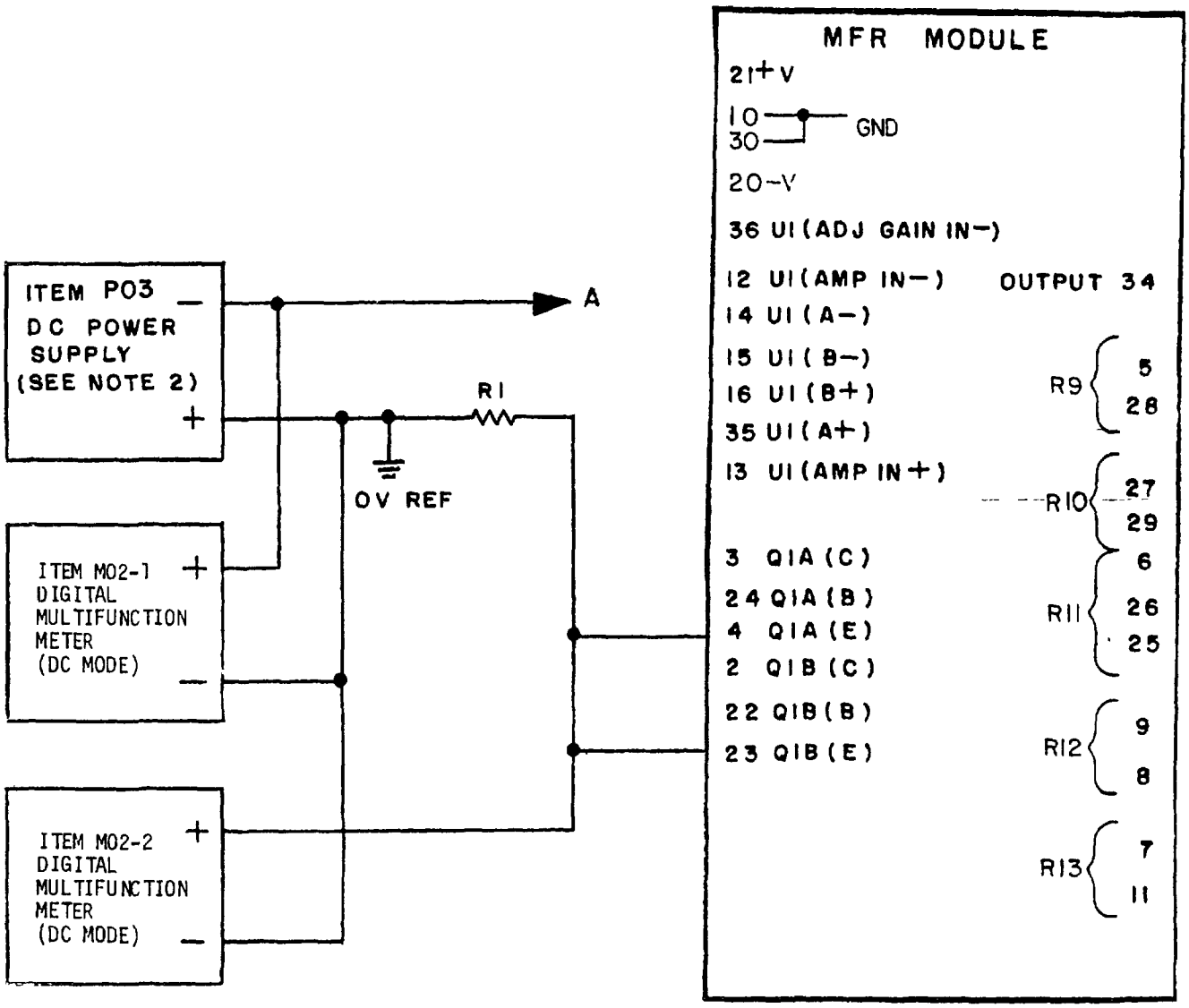
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NOTES:
1. R1 = 500 kΩ ±1%, 1/8 W.
2. The test shall be performed as follows:
   a. Initially set item P07 to 0 V.
   b. Gradually increase the voltage on item P07 until either the specified current limit is reached or 5 V is reached on item M02 (Do not exceed 5 V).
   c. The module passes if 5 V is reached before the specified current limit is reached (Do not exceed the maximum allowable current limit). Record the item M03 reading for each test condition.

FIGURE 21. Breakdown voltage ($V_{(BR)EBO}$) check test circuit for table XVI.

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NOTES:
1. R1 = 100Ω ±1%, 1/4 W.
2. The test shall be performed as follows:
   a. Initially set item P03 output to 0 V.
   b. Connect point A as indicated on table XVII and gradually increase the output of item P03 until either the specified voltage limit of table XVII is reached on item M02-2 or -61 V is reached on item M02-1 (Do not exceed the specified voltage limit on item M02-2 or -61 V on item M02-1).
   c. The module passes if -61 V is reached on item M02-1 before the specified voltage limit is reached on item M02-2. Record the item M02-2 voltage reading for each test condition.

FIGURE 22. Breakdown voltage ($V_{(BR)CEO}$) test circuit for table XVII.

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